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Studierichting: Informatica  
Oefeningenreeks 5A nr. 11.4

$$\begin{aligned} & \int_0^{-1} e^x(x^2 + 3x + 2)dx \\ &= \int_0^{-1} e^x x^2 dx + 3 \int_0^{-1} e^x x dx + 2 \int_0^{-1} e^x dx \end{aligned}$$

Eerst berekenen we:

$$\int e^x x^2 dx$$

$$u = x^2$$

$$du = 2x dx$$

$$dv = e^x dx$$

$$v = e^x$$

$$\int e^x x^2 dx = x^2 e^x - \int 2x e^x dx$$

Nu berekenen we:

$$\int 2x e^x dx$$

$$u = 2x$$

$$du = 2 dx$$

$$dv = e^x dx$$

$$v = e^x$$

$$\begin{aligned} \int 2x e^x dx &= 2x e^x - \int 2e^x dx \\ &= 2x e^x - 2e^x + C_1 \end{aligned}$$

Dus:

$$\begin{aligned} \int e^x x^2 dx &= x^2 e^x - (2x e^x - 2e^x) + C_2 \\ &= x^2 e^x - 2x e^x + 2e^x + C_2 \\ &= e^x(x^2 - 2x + 2) + C_2 \end{aligned}$$

Verder:

$$\int e^x x dx$$

$$u = x$$

$$du = dx$$

$$dv = e^x dx$$

$$v = e^x$$

$$\int e^x x dx = xe^x - \int e^x dx$$

$$= xe^x - e^x + C_3$$

$$= e^x(x - 1) + C_3$$

Ook:

$$\int e^x dx = e^x + C_4$$

Nu invullen in de bepaalde integraal:

$$\begin{aligned} & \int_0^{-1} e^x(x^2 + 3x + 2)dx \\ &= [e^x(x^2 - 2x + 2) + 3e^x(x - 1) + 2e^x]_0^{-1} \\ &= [e^x(x^2 - 2x + 2 + 3x - 3 + 2)]_0^{-1} \\ &= [e^x(x^2 + x + 1)]_0^{-1} \\ &= e^{-1}((-1)^2 + (-1) + 1) - e^0(0^2 + 0 + 1) \\ &= e^{-1}(1 - 1 + 1) - 1 \\ &= e^{-1} - 1 \\ &= \frac{1}{e} - 1 \end{aligned}$$

## References